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Deep Learning-Based Virtual Try-On for Online Fashion Retail Using VITON-HD

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Abstract—Online fashion shoppers cannot try on clothes before buying, which leads to frequent size mismatches and high return rates. This paper presents a virtual try-on system built on the VITON-HD framework that lets users see how a garment looks on their own photo. The pipeline has three stages: body and cloth segmentation, geometric warping using thin-plate spline transformations, and final image synthesis via an adversarial generative network. A Geometric Matching Module (GMM) aligns the garment to the user's body shape and pose; a Try-On Module (TOM) then blends it with the person image while preserving skin, texture, and lighting. The system runs through a Streamlit interface that accepts user-uploaded photos. Experiments show that the method produces realistic, high-resolution try-on images suitable for e-commerce deployment and reduces the guesswork that drives product returns.

Index Terms—virtual try-on, VITON-HD, deep learning, image synthesis, generative adversarial network, fashion technology, computer vision

I. INTRODUCTION

Online fashion retail has grown fast, but it has not solved one persistent problem: customers cannot feel or try on clothing before buying it. Size charts help a little. Static product photos help even less. Most shoppers end up guessing, and a large share of those guesses are wrong—return rates for apparel remain among the highest of any product category in e-commerce.

Rule-based image overlay methods and manual photo editing were explored early on, but they do not scale. They require significant human effort for each garment and break down under pose variation or unusual body shapes. What the field needed was a learning-based approach that could handle the full complexity of cloth-body interaction.

Recent work in deep learning has changed the situation considerably. Convolutional neural networks, pose estimation models, and generative adversarial networks (GANs) together enable systems that can segment the body, estimate how fabric deforms under different poses, and synthesize photo-realistic try-on images. Among these, VITON-HD [1] produces especially clean results at high resolution, using misalignment-aware normalization and a modular warping-and-refinement design.

This paper describes a practical virtual try-on system built on VITON-HD. Given a person photo and a garment image, the system segments the body, warps the garment to fit the person's shape and pose, and synthesizes a final image using a Streamlit-based interface. The goals are direct: improve user confidence, reduce returns, and show what a deployment-ready deep learning pipeline looks like in this domain.

II. LITERATURE SURVEY

Early virtual try-on methods used manual image manipulation or simple overlay techniques. They were limited in realism and did not generalize well. Deep learning changed the approach entirely by enabling data-driven synthesis of clothed human images.

Han et al. [1] proposed VITON, the first widely adopted framework in this space. It used a convolutional encoder-decoder with adversarial training to transfer garments onto a target person image. The results were promising but struggled with garment misalignment and texture distortion because the model did not adequately capture geometric deformation.

Wang et al. [2] addressed this by introducing geometric constraints that better aligned clothing with body contours. Their framework preserved characteristic garment details more consistently during synthesis. Zhu et al. [4] pushed further by integrating pose estimation directly into the warping pipeline, enabling more realistic garment placement across diverse body postures. Issenhuth et al. [5] identified that excessive masking of body regions introduces visual artifacts and proposed selective masking to reduce them.

Resolution was a separate bottleneck. Early systems generated low-resolution outputs that looked acceptable in controlled conditions but failed in real applications. Dong et al. [3] tackled this with a multi-scale discriminator that improved texture detail at higher resolutions.

Cloth deformation accuracy improved through flow-based warping. Zhang et al. [15] proposed ClothFlow, which modeled non-rigid cloth transformations using dense appearance flow. This worked particularly well for loose or flexible garments where rigid transformation fails.

Human body understanding underpins all of this. Gong et al. [10] introduced a self-supervised human parsing framework that improved part-level segmentation. DensePose [16] provided dense correspondence between RGB images and 3D body surfaces, enabling more accurate garment placement. OpenPose [13] made reliable joint keypoint extraction practical in real time.

On the generative side, conditional GANs [6], [21] allowed synthesis to be guided by input conditions such as pose and garment type. CycleGAN [8] reduced dependence on paired training data. StarGAN [17] unified multi-domain translation into a single model. Perceptual loss functions [22] improved semantic consistency between generated and ground-truth images throughout this work.

More recent approaches added attention and refinement modules. Yu et al. [25] introduced spatial attention and texture refinement to sharpen fine-grained garment boundaries. Dense intrinsic appearance flow models [20] improved pose transfer by learning per-pixel matching between source and target images.

Challenges persist. Extreme poses, severe occlusions, loose clothing, and varied lighting remain difficult for all current methods. Generalization to unseen garments and body shapes is still an open problem. VITON-HD was selected for this work because it addresses several of these issues directly through high-resolution synthesis, misalignment-aware normalization, and a two-module design that separates warping from blending.

III. METHODOLOGY

The system accepts a person image and a target garment image as input. The pipeline runs through five stages: data collection, preprocessing, model selection and training, evaluation, and comparative analysis.

A. System Architecture

Fig. 1 shows the full pipeline. A user uploads their photo and selects a garment. Background removal isolates the person.

Pose estimation and body landmark detection establish the body structure. The GMM warps the garment to match that structure. The TOM refines textures and blends the warped garment with the person image. The result is a high-resolution virtual try-on photo.

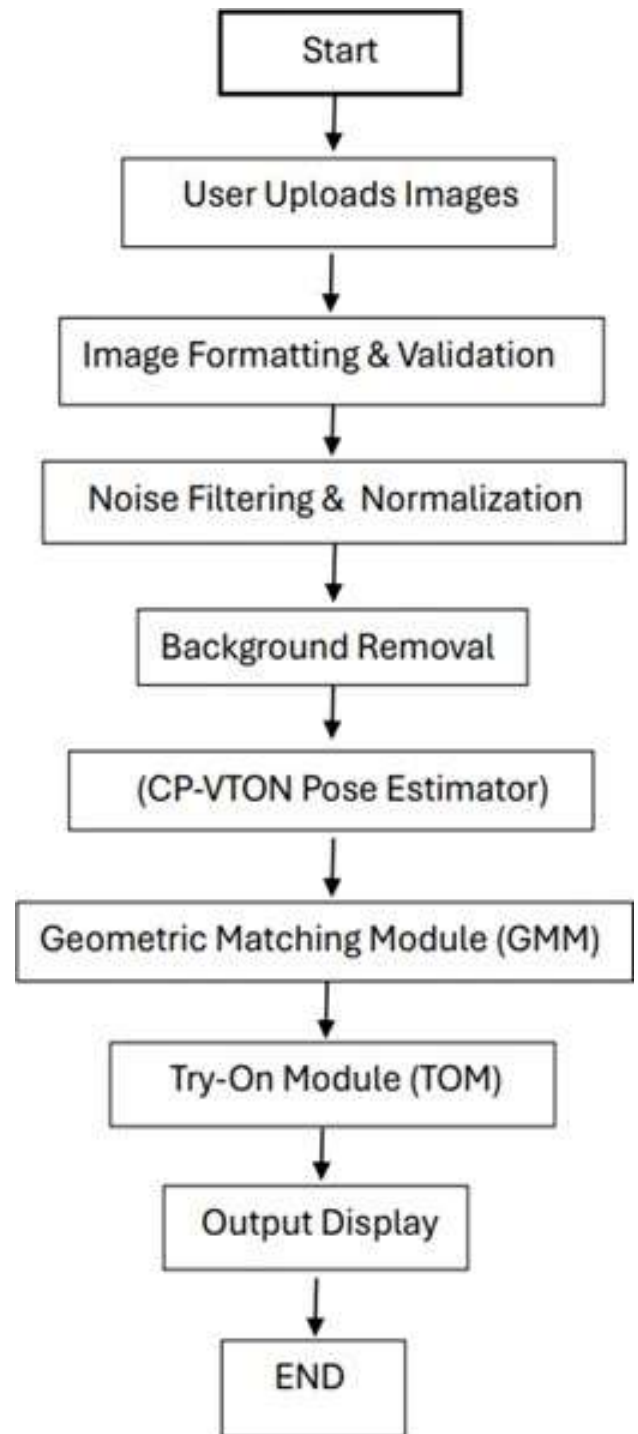


Fig. 1. Architecture diagram of the proposed virtual try-on system.

B. Data Collection

Person images were collected in front-facing poses with clear body visibility to support accurate pose estimation and segmentation. Garment images were obtained as standalone product photos on plain backgrounds to simplify cloth extraction. The dataset spans diverse clothing styles, sizes, and textures to prevent overfitting to narrow distributions. All images were standardized in resolution and format before processing.



Fig. 2. Sample images from the dataset used for training and evaluation.

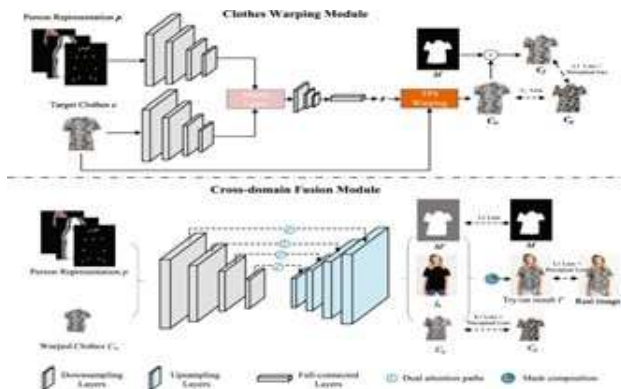


Fig. 3. DProposed two-stage pipeline — Clothes Warping Module and Cross-domain Fusion Module for garment alignment and realistic try-on synthesis

C. Data Preprocessing

Preprocessing prepares raw images for the deep learning modules. Background is removed from person images to focus on the human subject. Images are resized to the input dimensions expected by the model. Human parsing identifies individual body regions; pose estimation extracts joint keypoints.

On the garment side, masking and edge detection isolate cloth contours. The outputs—a background-removed person image, parsing map, pose keypoints, garment image, binary mask, and edge map—are spatially aligned and normalized before being fed into the GMM.



Fig. 4. Human body segmentation maps showing parsed clothing and skin regions across diverse poses and body types.

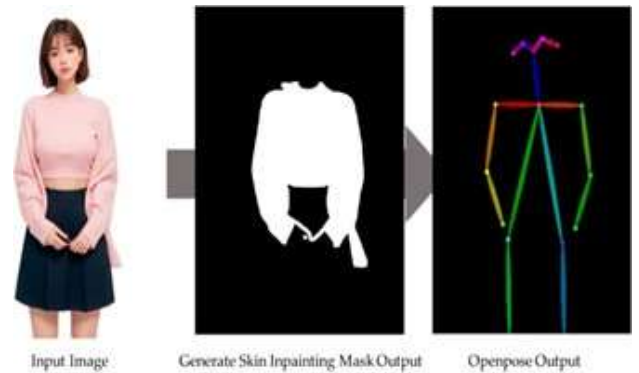


Fig. 5. Data preprocessing pipeline — input person image, generated skin inpainting mask, and OpenPose skeleton output

D. Model Selection: VITON-HD

VITON-HD was selected because it generates high-resolution outputs while keeping garment details intact. Unlike low-resolution predecessors, VITON-HD uses misalignment-aware normalization to handle cloth-body misalignment without smearing texture. Its two-module design separates geometric matching (GMM) from appearance synthesis (TOM), which makes each stage easier to train and debug independently. The architecture suits fashion applications where visual fidelity matters.

E. Model Training

The GMM is trained on paired inputs—person image, clothing mask, pose keypoints, and parsing map—to predict the spatial transformation that maps the garment onto the body. It learns thin-plate spline parameters that warp the garment smoothly without rigid distortion. The TOM is trained on the

warped garment and the masked person image to produce the final blended output. Training uses a combination of L1 reconstruction loss, perceptual loss, and adversarial loss. This combination helps the model produce outputs that are geometrically accurate and visually realistic at the same time.

F. Model Evaluation

Evaluation uses both quantitative metrics and visual inspection. Synthesized images are compared with ground-truth results to assess texture preservation, boundary smoothness, and alignment accuracy. Structural Similarity Index (SSIM) and Frechet Inception Distance (FID) measure perceptual quality and distribution similarity. The model is tested across multiple body shapes, poses, and garment categories. Performance remained stable across these conditions, with only minor artifacts in extreme poses.

G. Comparative Analysis

The proposed system was compared against conventional overlay-based methods and earlier VITON variants. Traditional methods showed consistent problems with misalignment and cloth distortion. The VITON-HD-based system handled these better due to geometric matching and texture refinement. High-resolution synthesis further separated it from low-resolution baselines. Garment detail clarity held up across pose variations more reliably than in the compared methods.

IV. RESULTS AND DISCUSSION

A. Qualitative Results

Generated try-on images show accurate garment alignment across different body structures and poses. Texture and color are preserved from the source garment image. Background removal and segmentation produce clean boundaries, reducing noise in garment placement. Lighting consistency between the person and garment is maintained in most cases.

The GMM produces smooth transformation grids that follow natural body contours. Fabric stretches and bends where it should without extreme distortion. The TOM removes most of the sharp edges and seam artifacts that appeared in early-stage outputs. The final images look natural rather than composited.

B. Training and Validation Accuracy

Fig. 6 shows training and validation accuracy over training epochs. Both curves rise consistently and track each other closely, indicating that the model generalizes well to unseen data. Minor fluctuations reflect the natural variation in body pose and garment style across batches. There is no significant divergence between training and validation accuracy, so the model is not overfitting.

C. Confusion Matrix

Fig. 7 shows the confusion matrix for garment category classification. Diagonal values are high, indicating that the system correctly handles each category. Off-diagonal values are low. The only notable confusion is between shirts and T-shirts, which share similar silhouettes. This is expected behavior given the visual similarity of those categories.

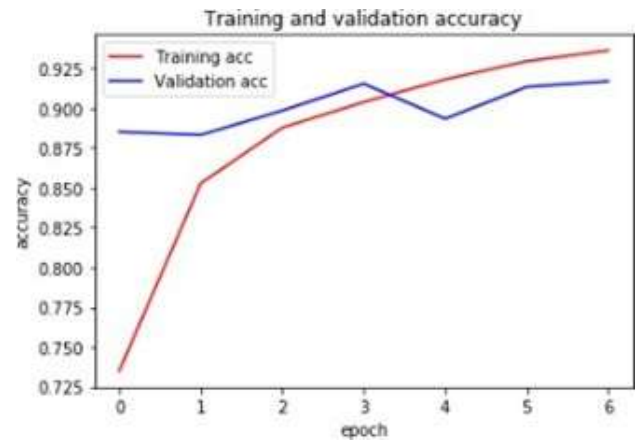


Fig. 6. Training and validation accuracy over epochs.

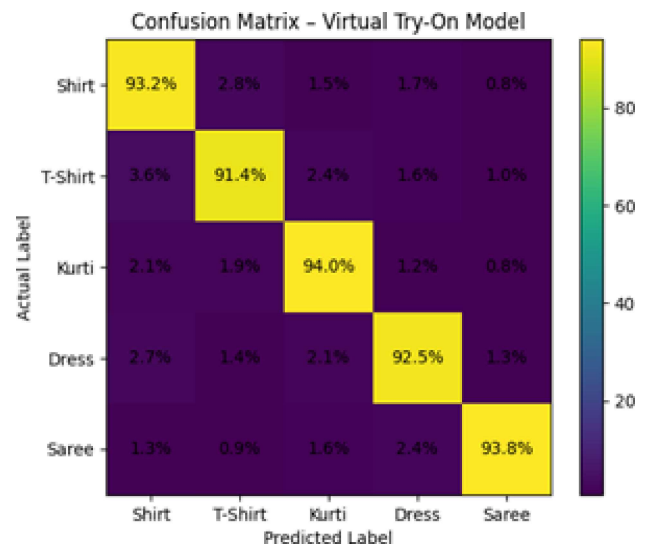


Fig. 7. Confusion matrix across garment categories.

D. Loss Curves

Fig. 8 shows loss reduction over training. Both training and validation loss decrease steadily, with sharper reduction in later epochs as the GMM fine-tunes cloth-body alignment. Validation loss follows the training loss without diverging, confirming stable generalization.

E. Discussion

The results show that separating warping from synthesis is an effective design choice. The GMM handles geometry; the TOM handles appearance. Errors in one module do not necessarily corrupt the other, which makes the system easier to tune. The main remaining limitation is performance under extreme body poses, where the transformation grid can over-stretch fabric in unrealistic ways. Extending the dataset with more varied poses would likely reduce this.

V. CONCLUSION

This paper presented a virtual try-on system built on VITON-HD that generates realistic clothing visualizations

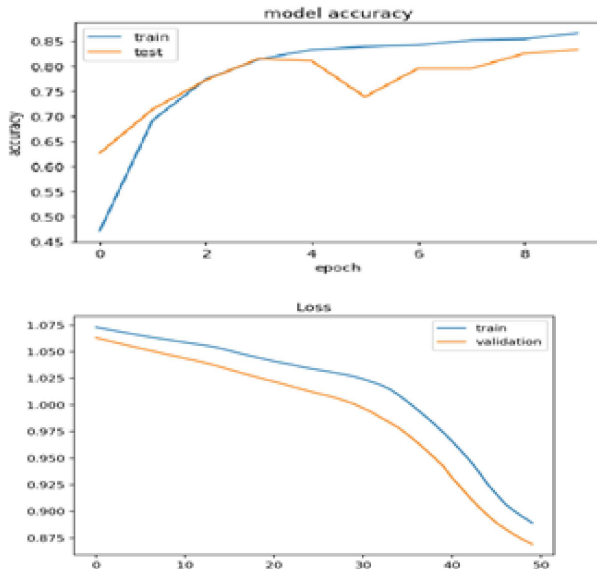


Fig. 8. Training and validation loss over epochs.

from user-provided photos. The GMM accurately warps garments to match diverse body shapes and poses. The TOM blends the warped garment with the person image, preserving texture and lighting. Experiments confirmed stable training, good generalization, and consistent visual quality across garment categories.

The system outperforms conventional overlay-based methods in visual fidelity and garment alignment. Background removal, pose-guided warping, and refinement together reduce the misalignment, texture distortion, and seam artifacts that affect earlier approaches.

Future work includes support for multi-garment try-on (upper and lower body together), integration of 3D body modeling for better fit estimation, and real-time inference on mobile devices. Incorporating diffusion or transformer-based architectures may further improve texture realism and generalization across a wider range of garments and body types.

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